How Organizational Inclusion Shapes Patient Satisfaction in Magnet Hospitals

Hyunmin Yu, PhD,†‡ Heather Brom, PhD,*† José A. Bauermeister, PhD,†‡ Jessie Reich, PhD,§ Matthew D. McHugh, PhD,*† Tari Hanneman, MPA,|| and Linda H. Aiken, PhD*†*

Objective: Magnet hospitals exhibit higher patient satisfaction than non-Magnet hospitals, yet the underlying mechanisms driving these differences remain underexplored. This study examined the associations between Magnet status, hospitals' inclusion efforts for diverse populations, and patient satisfaction, and whether inclusion efforts explain Magnet hospitals' higher satisfaction.

Methods: This cross-sectional study analyzed 2023 secondary data from 4 sources: the Hospital Consumer Assessment of Healthcare Providers and Systems (HCAHPS), the Healthcare Equality Index (HEI), the American Hospital Association Annual Survey, and the list of Magnet-recognized organizations. The sample included 708 hospitals (216 Magnet, 492 non-Magnet) participating in both HCAHPS and HEI. HEI scores, which assess hospitals' inclusion efforts for lesbian, gay, bisexual, transgender, queer, and other sexual and gender diverse (LGBTQ+) populations, were used as a proxy for overall inclusion. Patient satisfaction was measured using 8 HCAHPS indicators. Mediation analyses tested whether HEI scores explained the association between Magnet designation and patient satisfaction.

Results: Magnet hospitals had higher HEI scores (M = 92.0, SD = 12.2) compared with non-Magnet hospitals (M = 88.5, SD = 13.3). They also had higher hospital ratings (M = 88.4, SD = 2.4 vs. M = 87.6, SD = 3.3) and patient recommendations (M = 88.4, SD = 3.2 vs. M = 86.8, SD = 4.1). Magnet status had direct effects on hospital ratings (b = 1.75, $P < 0.001$) and recommendations (b = 2.37, $P < 0.001$), as well as indirect effects through HEI performance on hospital ratings (b = 0.07, $P = 0.022$) and recommendations (b = 0.10, $P = 0.026$), resulting in total effects on hospital ratings (b = 1.82, $P < 0.001$) and recommendations (b = 2.47, $P < 0.001$).

Conclusions: The findings underscore the importance of organizational priorities and policies that promote patient-centeredness and inclusion for the satisfaction of all patients.

Key Words: organizational inclusion, patient-centeredness, Magnet hospitals, patient satisfaction, Healthcare Equality Index

(*Med Care* 2025;00:000–000)

Patient satisfaction is a key indicator of hospital performance, linked to care quality, clinical outcomes, and hospital reputation.[1–3] The Hospital Consumer Assessment of Healthcare Providers and Systems (HCAHPS) survey is the primary tool for measuring patient satisfaction in US hospitals. It enables national comparisons and ties patient satisfaction scores to hospitals reimbursement through the Value-Based Purchasing program for hospitals under the Inpatient Prospective Payment System (IPPS), directly impacting financial outcomes.[4] Understanding the hospital features that contribute to better HCAHPS performance is important for clinicians, health system leaders, policymakers, and payers.

One such organizational factor may be inclusion. Management theories on inclusive organizations define *inclusion* as fostering an organizational culture where individuals of all backgrounds are valued, fairly treated, and provided equal opportunities to participate, contribute, and advance. Beyond representation, inclusion emphasizes meaningful involvement, equitable access to decision-making, and integration of diverse identities into the organizational fabric, ensuring a sense of belonging for all employees.[5] By fostering inclusion, organizations can improve their culture, enhance employee well-being, and elevate overall organizational performance, including customer satisfaction.[5–7] Studies suggest that inclusive workplaces are associated with improved nurse well-being, including lower burnout and job dissatisfaction, as well as

Received for publication March 17, 2025; accepted July 17, 2025.
From the *Center for Health Outcomes and Policy Research, School of Nursing, University of Pennsylvania, Philadelphia, Pennsylvania; †Leonard Davis Institute of Health Economics, University of Pennsylvania, Philadelphia, Pennsylvania; ‡Eidos LGBTQ+ Health Initiative, School of Nursing, University of Pennsylvania, Philadelphia, Pennsylvania; §Hospital of the University of Pennsylvania, Philadelphia, Pennsylvania; and ||Human Rights Campaign Foundation, Washington, DC.
This research was supported by the National Institute of Nursing Research, National Institutes of Health (R01NR014855). The primary author received postdoctoral fellowship funding from the Eidos LGBTQ+ Health Initiative at the University of Pennsylvania. The funders had no role in the study's design and conduct, including data collection, management, analysis, and interpretation. The funders were not involved in the preparation, review, or approval of the manuscript, nor in the decision to submit it for publication.
The authors declare no conflict of interest.
Correspondence to: Hyunmin Yu, PhD, School of Nursing, University of Pennsylvania, 418 Curie Boulevard, Philadelphia, Pennsylvania 19104. E-mail: hyuy@nursing.upenn.edu.
Supplemental Digital Content is available for this article. Direct URL citations are provided in the HTML and PDF versions of this article on the journal's website, www.lww-medicalcare.com.
Copyright © 2025 Wolters Kluwer Health, Inc. All rights reserved.
DOI: 10.1097/MLR.0000000000002201

Copyright © 2025 Wolters Kluwer Health, Inc. Unauthorized reproduction of this article is prohibited.

higher-quality care[8] and greater patient satisfaction.[9]

Magnet hospitals consistently outperform non-Magnet hospitals on HCAHPS scores,[10–13] potentially driven by enhanced nurse inclusion and a commitment to patient-centered care. Magnet hospitals are recognized by the American Nurses Credentialing Center (ANCC) for excellence in nursing practice, patient care, and innovation.[14] Although the Magnet framework emphasizes diversity, equity, and inclusion,[15] it does not explicitly define measures for inclusivity for specific populations such as lesbian, gay, bisexual, transgender, queer or questioning, and other sexual and gender-diverse (LGBTQ+) individuals.

Magnet hospitals' commitment to patient-centered care is demonstrated through their focus on evidence-based nursing care, interdisciplinary collaboration, clinician inclusion in decision-making, and policies that prioritize patient needs and experiences.[16] For example, during the HIV/AIDS epidemic, Magnet hospitals were shown to have lower AIDS-related mortality and higher patient satisfaction with AIDS-related care than comparison hospitals, even those with specialized AIDS units.[17] Thus, Magnet hospitals, with their strong patient-centered focus, have likely been more inclusive of diverse patient populations, which may have contributed to their consistently high patient satisfaction scores.

There is a lack of comprehensive measures to assess hospitals' inclusion efforts for diverse patient populations. One of the few existing metrics is the Healthcare Equality Index (HEI). Since 2007, the Human Rights Campaign Foundation (HRCF) has evaluated US health care facilities that voluntarily participate in its survey, assessing policies and practices that promote LGBTQ+ inclusion, including nondiscrimination policies, inclusive services, employee benefits, and community engagement.[18] While the HEI specifically focuses on LGBTQ+ inclusion, it may serve as a useful proxy for broader inclusivity efforts. Hospitals that implement policies to support one marginalized population may be more likely to adopt inclusive practices for other historically underserved groups. Guided by theories on inclusive organizations,[5–7] this study explores associations between Magnet status, HEI performance, and patient satisfaction. We hypothesized that Magnet status is directly associated with greater patient satisfaction and that Magnet status is indirectly associated with higher patient satisfaction through higher levels of HEI performance (Fig. 1).

METHODS

Study Design and Data Sources

This cross-sectional, observational study integrated hospital data from 4 sources collected in 2023, including the HEI, HCAHPS, American Hospital Association (AHA) surveys, and the list on Magnet hospitals. The HCAHPS measures patient satisfaction with hospital care through random sampling of discharged patients.[19] HCAHPS data were accessed from the Centers for Medicare & Medicaid Services (CMS) public reporting website.[20] The HEI evaluates the LGBTQ+ inclusivity of health care policies and practices based on a voluntary, detailed online survey with supporting documentation for verification,[18] assigning scores from 0 to 100 that are publicly available on its website.[21] The complete HEI dataset was obtained from the HRCF. The AHA Annual Survey provides insights into hospital characteristics, staffing, and operations.[22] Although the AHA dataset is not publicly available, it contains only hospital-level information without any patient-level data or protected health information (PHI). Access to the AHA dataset was obtained directly from the AHA. In addition, the ANCC maintains updated lists of Magnet hospitals, which were obtained from their public reporting website.[23]

Data linkage was performed using unique hospital identifiers. First, the AHA hospital identifier was used to link the HEI dataset to the AHA dataset. Next, the Medicare number was used to link the HCAHPS dataset to the merged AHA-HEI dataset. Finally, each hospital in the merged dataset was manually categorized by Magnet status using the ANCC's list of Magnet hospitals. All datasets used in this study are either publicly available with hospital identifiers or contain only hospital-level data without patient information. Given the absence of PHI and the exclusive use of de-identified hospital-level data, this study received an exemption from Institutional Review Board review.

Study Sample

To evaluate the direct relationship between hospitals' Magnet status and HCAHPS scores, as well as the potential indirect association through HEI performance, we applied the following inclusion criteria: (1) hospitals located in the continental United States, (2) participation in the 2023 HCAHPS survey with reported scores, and (3) participation in the HEI assessment in 2023.

We initially included 4779 hospitals that participated in the HCAHPS survey in 2023. The CMS calculates and reports linear mean scores for HCAHPS measures to facilitate robust hospital comparisons. However, these scores are not reported for hospitals with fewer than 100 completed surveys over a 4-quarter period to ensure reliability. As a result, we excluded 1572 hospitals with no reported scores. We then excluded 2497 hospitals that did not participate in the HEI in 2023. In addition, given the substantial differences in health care policies between US territories and the 50 states,[24] we excluded 2 hospitals located in US territories. Our final analytic sample for the primary analysis included 708 hospitals, including 216 Magnet hospitals and 492 non-Magnet hospitals (Fig. 2). The sensitivity analysis focused exclusively on large-sized hospitals (>250 beds), which demonstrated the largest differences in hospital characteristics by Magnet status, including 158 Magnet hospitals and 187 non-Magnet hospitals.

Measures

Dependent Variable: Patient Satisfaction With Hospital Care

Our outcome was patient satisfaction with hospital care. Among the 10 publicly reported HCAHPS measures,[25] we excluded 2 that are not directly related to

Copyright © 2025 Wolters Kluwer Health, Inc. All rights reserved.

Copyright © 2025 Wolters Kluwer Health, Inc. Unauthorized reproduction of this article is prohibited.

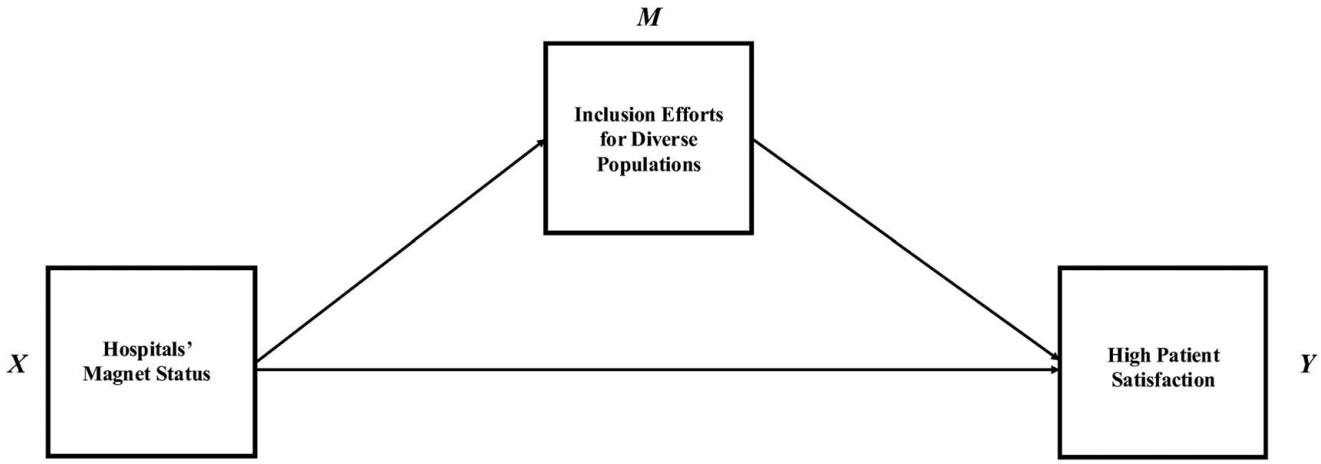

FIGURE 1. Conceptual Diagram. This diagram hypothesizes a direct positive association between a hospital's Magnet status and patient satisfaction, as well as an indirect association through the hospital's inclusion efforts for diverse populations.

patient care: cleanliness of the hospital environment and quietness of the hospital environment. Thus, we used 2 global measures, including overall hospital rating and patients' recommendation of the hospital, as primary outcomes, and 6 specific domain measures, including communication with nurses, communication with doctors, responsiveness of hospital staff, communication about medicines, discharge information, and care transition, as secondary outcomes. A detailed description of the questions and response options for each of the 8 outcome measures can be found in Supplemental Digital Content 1, http://links.lww.com/MLR/D21.

CMS calculates hospitals' linear mean scores for these outcomes, ranging from 0 to 100, using a multistep approach: (1) numeric values are assigned to individual responses and averaged for each hospital over a 12-month period (from January 2023 to December 2023 in our study); (2) these averages are adjusted for patient characteristics (eg, age, education, self-rated health, service line, and primary language spoken at home) and scaled to a 0–100 range; (3) the adjusted scores are further refined based on survey mode; and (4) the final scores are weighted according to the number of eligible patients treated by the hospital in each quarter of the reporting period.[25]

Independent Variable: Magnet Designation Status

In line with our analysis of HCAHPS and HEI surveys collected in 2023, this variable was a binary variable, with a value of 1 if they held Magnet status in 2023 and 0 if they did not.

Mediating Variable: Performance in the HEI

We used overall HEI scores as a proxy for hospitals' inclusion efforts for diverse populations and as a potential mediator between Magnet status and patient satisfaction. The total HEI score (0–100) comprises 4 subcategories: (1) *Nondiscrimination and Staff Training* (up to 40 points) assesses nondiscrimination policies for patients and staff, along with staff training for LGBTQ+inclusive care; (2) *Patient Services and Support* (up to 30 points) evaluates the availability of LGBTQ+specific clinical services; (3) *Employee Benefits and Policies* (up to 20 points) measures support for LGBTQ+ employees, such as inclusive health insurance coverage; and (4) *Patient and Community Engagement* (up to 10 points) reflects outreach efforts and engagement with the LGBTQ+ community.[26] We used total HEI scores for primary analyses and the 4 HEI subscores for sensitivity analyses.

Covariates: Hospital Characteristics

We included structural, operational, and geographical hospital-level covariates that have been shown to be associated with patient satisfaction, Magnet status, and HEI performance. These covariates included hospital size, teaching status, specialized service capability, ownership, region, metropolitan location, system affiliation, and participation in the Medicare Inpatient Prospective Payment System (IPPS).

Teaching status was determined by the ratio of resident physicians and fellows to hospital beds. Hospitals were classified as nonteaching, minor teaching (ratio of $\leq$1:4), or major teaching (ratio $>$1:4).[27] Specialized service capability was categorized based on the availability of advanced procedures. Hospitals offering open-heart surgery or major transplants were classified as having high capability, while those without these services were considered to have low capability.[28] Hospital size was categorized as small ($\leq$100 beds), medium (101–250 beds), or large ($>$250 beds).[29] Metropolitan status was assigned using the Core-Based Statistical Area system,[30] classifying hospitals as either metropolitan or nonmetropolitan, with rural and micropolitan areas grouped into the nonmetropolitan category. Hospital ownership was categorized as federal, for-profit, or not-for-profit, with the latter including both nonfederal government and nongovernmental entities. System membership was coded as a binary variable, including hospitals that were system-

Copyright © 2025 Wolters Kluwer Health, Inc. All rights reserved.

Copyright © 2025 Wolters Kluwer Health, Inc. Unauthorized reproduction of this article is prohibited.

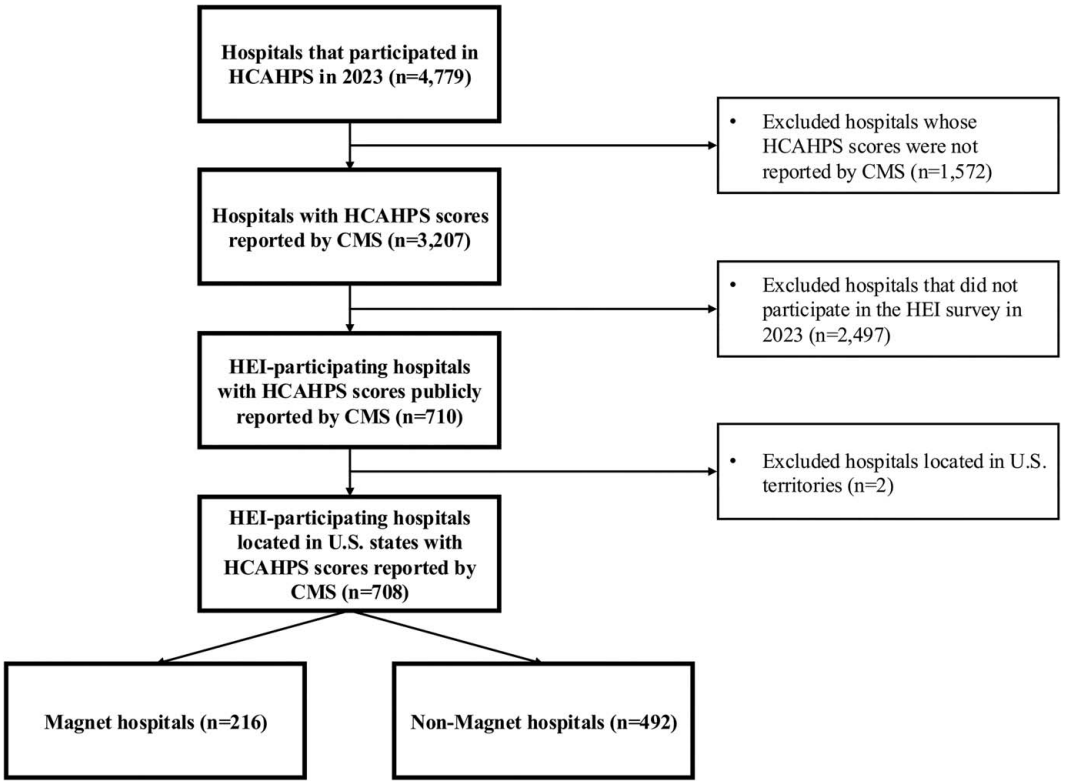

FIGURE 2. Hospital Sample Selection Flowchart. Abbreviations: CMS = Centers for Medicare & Medicaid Services; HCAHPS = Hospital Consumer Assessment of Healthcare Providers and Systems; HEI = Healthcare Equality Index. Sample selection flowchart showing inclusion and exclusion criteria for hospitals in the analytic sample. Hospitals were included if they participated in the HEI in 2023 and had HCAHPS scores publicly reported by CMS in 2023. Exclusions were based on HEI participation, HCAHPS reporting status, and location in US territories.

affiliated and those that were not. Participation in the IPPS was also coded as a binary variable, with hospitals classified as either participating or nonparticipating (eg, critical access hospitals and children's hospitals).

Statistical Analysis

All data analyses were conducted using R version 4.4.1. Using bivariate analyses, we examined differences in patient satisfaction outcomes, HEI performance, and hospital characteristics by Magnet status. To assess the direct relationship between Magnet designation and patient satisfaction, as well as the potential indirect association through HEI performance, we conducted mediation analysis using the *mediation* package in R.[31,32]

In this analysis, the total association of Magnet designation had 2 components: (1) the mediation (indirect) association, which indicates how much of the association of Magnet designation on patient satisfaction is explained by the mediator (HEI performance), and (2) the direct association, which reflects Magnet designation's association independent of the mediator.[33,34] We estimated the average mediation (indirect) associations and average direct associations.[32] The average mediation (indirect) associations indicate the average extent to which the mediator (HEI performance) accounts for the association of Magnet designation on patient satisfaction, quantifying the improvement in patient satisfaction attributed to changes in the mediator. In contrast, the average direct associations represent the average association of Magnet designation on patient satisfaction that occurs without relying on HEI performance.

We fit 8 separate models for the 8 different patient satisfaction outcome measures. Linear regression with least squares was used for the mediator model, with HEI scores as the dependent variable and Magnet status and covariates as independent variables. Next, we modeled HCAHPS scores as the dependent variable using linear regression with least squares, including Magnet status, HEI scores, and the same covariates as independent variables. The *mediate* function from the R package was then used to estimate the average mediation (indirect) and direct associations. A nonparametric bootstrap approach with 1000 replications and bias-corrected and accelerated intervals was applied, as recommended for estimating mediation (indirect) associations.[35]

RESULTS

Table 1 summarizes hospital characteristics, HEI scores, and HCAHPS scores by Magnet status. Of the 708 hospitals in the study, 216 (30.5%) were Magnet hospitals,

Copyright © 2025 Wolters Kluwer Health, Inc. All rights reserved.

Copyright © 2025 Wolters Kluwer Health, Inc. Unauthorized reproduction of this article is prohibited.

while 492 (69.5%) were non-Magnet. Compared with non-Magnet hospitals, Magnet hospitals were more likely to be in metropolitan areas (99% vs. 87%), be larger institutions (73% vs. 38%), have high specialized service capability (64% vs. 29%), operate as not-for-profit hospitals (98% vs. 82%), participate in the IPPS (97% vs. 79%), and be teaching hospitals (62% vs. 55%).

Magnet hospitals had higher HEI scores (M = 92.0, SD = 12.2) compared with non-Magnet hospitals (M = 88.5, SD = 13.3; $P = 0.001$). They also had higher HCAHPS scores for 2 global measures: overall hospital rating (M = 88.4, SD = 2.4 vs. M = 87.6, SD = 3.3; $P = 0.001$) and likelihood to recommend the hospital (M = 88.4, SD = 3.2 vs. M = 86.8, SD = 4.1; $P < 0.001$). In addition, they reported higher satisfaction with care transition (M = 81.3, SD = 2.0 vs. M = 80.7, SD = 2.9; $P = 0.011$).

Primary Analyses

Magnet status had both statistically significant direct and indirect associations with 2 global patient satisfaction measures (Table 2). Magnet status was directly associated with higher hospital ratings (b = 1.75, $P < 0.001$) and greater likelihood among patients to recommend their hospitals (b = 2.37, $P < 0.001$). In addition, Magnet status had an indirect positive association with hospital ratings (b = 0.07, $P = 0.022$) and likelihood to recommend (b = 0.10, $P = 0.026$) through higher HEI performance. HEI performance accounted for 4% of the overall relationship between Magnet status and both global patient satisfaction measures.

Magnet status had statistically significant direct associations with all 6 specific domain measures and statistically significant indirect associations with 4 out of 6 domain measures through higher HEI performance (Table 3). HEI performance accounted for 4% of the relationship between Magnet status and satisfaction with doctors' communication and medication communication, respectively. In addition, HEI performance accounted for 3% of the relationship between Magnet status and satisfaction with staff responsiveness and care transition, respectively.

Sensitivity Analyses

Sensitivity analyses using 4 HEI subscores revealed that Magnet status had statistically significant indirect associations with all 8 patient satisfaction measures through the higher HEI subscore related to inclusive clinical patient services. However, subscores related to nondiscrimination policies, employee benefits, and community engagement were not significant mediators. Sensitivity analyses limited to large hospitals did not reveal differences in overall statistical significance.

DISCUSSION

Our study provides evidence that Magnet hospitals consistently achieve higher patient satisfaction scores compared with non-Magnet hospitals. In addition, our findings suggest that Magnet hospitals' commitment to inclusivity, as captured by higher HEI scores, may be one of the explanations of their higher patient satisfaction. Although the effect sizes are modest, even small increases in these weighted HCAHPS scores can result in substantial financial benefits, highlighting the significance of incremental improvements. Magnet hospitals also exhibited lower variability in these scores, indicating a higher likelihood of qualifying for incentives and a reduced risk of penalties. These results underscore the importance of organizational priorities and policies that promote patient-centeredness and inclusion for diverse populations in health care settings.

Consistent with prior research,[10–12] our study reinforces that Magnet designation is positively associated with patient satisfaction. Our study uniquely contributes evidence that HEI scores as a proxy for hospitals' inclusion efforts for diverse populations partially explain this relationship, suggesting that inclusivity may play a key role in shaping patient satisfaction. This may be because Magnet hospitals have enculturated throughout the organization a commitment to inclusion reflected in their commitment to patient-centered care for all patients. Our sensitivity analyses revealed that among various factors considered, inclusive clinical services designed to meet the needs of diverse patient populations were the only significant explanation for the association between Magnet status and all patient satisfaction measures. This finding suggests that patient-centeredness is not just a complementary aspect of Magnet hospitals' organizational priorities but a fundamental component of their inclusion efforts.

Another potential mechanism underlying this indirect association involves the impact of an inclusive organizational culture on health care workers. Management theories on inclusive organizations suggest that fostering an environment of inclusivity not only enhances employee well-being but also improves job satisfaction.[5–7] Given that Magnet hospitals have been recognized for prioritizing nurse well-being and demonstrating better nurse job outcomes compared with non-Magnet hospitals,[36–38] their commitment to inclusion, may further amplify these benefits. In turn, a more satisfied and supported health care workforce is likely to deliver higher-quality, patient-centered care that meets the needs of diverse populations. This emphasizes the potential impact of inclusion efforts on both workforce well-being and patient outcomes.

Although the indirect association through HEI scores was statistically significant, it accounted for only a small portion of the total association, highlighting the multifaceted nature of factors influencing patient satisfaction. Nevertheless, since HCAHPS ratings contribute 25% of a hospital's Value-Based Purchasing (VBP) Total Performance Score, which directly impacts Medicare reimbursements, hospitals with higher HCAHPS scores are positioned to receive greater financial rewards. This financial incentive further underscores the importance of implementing strategies that promote inclusion efforts.

Several limitations should be acknowledged. First, our cross-sectional design limits the ability to infer causality between Magnet status, HEI scores, and patient

Copyright © 2025 Wolters Kluwer Health, Inc. All rights reserved.

Copyright © 2025 Wolters Kluwer Health, Inc. Unauthorized reproduction of this article is prohibited.

TABLE 1. Hospital Characteristics and Outcomes Stratified by Magnet Status

	Total hospitals (n = 708)	Magnet hospitals (n = 216)	Non-Magnet hospitals (n = 492)	*P*
Total HEI score [mean (SD)]	89.5 (13.1)	92.0 (12.2)	88.5 (13.3)	0.001
Number of HCAHPS surveys in 2023 [mean (SD)]	953.3 (926.1)	1419.2 (1255.8)	748.7 (637.9)	< 0.001
HCAHPS scores				
Overall hospital rating [mean (SD)]	87.8 (3.1)	88.4 (2.4)	87.6 (3.3)	0.001
Overall hospital rating (range)	74–96	81–96	74–96	
Patients' recommendation [mean (SD)]	87.3 (3.9)	88.4 (3.2)	86.8 (4.1)	< 0.001
Patients' recommendation (range)	68–97	79–97	68–97	
Nurse communication [mean (SD)]	90.8 (2.2)	90.7 (2.5)	91.0 (1.6)	0.060
Nurse communication (range)	82–97	86–96	82–97	
Doctor communication [mean (SD)]	90.6 (2.0)	90.7 (1.5)	90.5 (2.2)	0.168
Doctor communication (range)	83–97	87–94	83–97	
Staff responsiveness [mean (SD)]	82.2 (4.3)	81.9 (2.9)	82.4 (4.7)	0.172
Staff responsiveness (range)	69–94	73–91	69–94	
Medication communication [mean (SD)]	75.7 (4.2)	75.4 (3.0)	75.8 (4.6)	0.299
Medication communication (range)	63–92	67–88	63–92	
Discharge information [mean (SD)]	86.2 (3.3)	86.5 (2.6)	86.1 (3.5)	0.173
Discharge information (range)	71–94	77–94	71–94	
Care transition [mean (SD)]	80.9 (2.7)	81.3 (2.0)	80.7 (2.9)	0.011
Care transition (range)	68–89	77–88	68–89	
Hospital size [n (%)]				< 0.001
Large (> 250 beds)	345 (48.7)	158 (73.1)	187 (38.0)	
Medium (101–250 beds)	221 (31.2)	47 (21.8)	174 (35.4)	
Small (≤ 100 beds)	142 (20.1)	11 (5.1)	131 (26.6)	
Teaching status [n (%)]				< 0.001
None	297 (41.9)	80 (37.0)	217 (44.1)	
Minor	192 (27.2)	84 (38.9)	108 (22.0)	
Major	219 (30.9)	52 (24.1)	167 (33.9)	
Specialized service capability [n (%)]				< 0.001
High	286 (40.4)	139 (64.4)	147 (29.9)	
Low	422 (59.6)	77 (35.6)	345 (70.1)	
Ownership [n (%)]				< 0.001
Not-for-profit	620 (87.6)	213 (98.6)	407 (82.7)	
Federal	76 (10.7)	3 (1.4)	73 (14.9)	
For-profit	12 (1.7)	0 (0.0)	12 (2.4)	
System affiliation [n (%)]				1.000
Affiliated	654 (92.4)	200 (92.6)	454 (92.3)	
Not affiliated	54 (7.6)	16 (7.4)	38 (7.7)	
IPPS participation [n (%)]				< 0.001
Yes	599 (84.6)	210 (97.2)	389 (79.1)	
No	109 (15.4)	6 (2.8)	103 (20.9)	
Metropolitan location [n (%)]				< 0.001
Metropolitan	643 (90.8)	215 (99.5)	428 (87.0)	
Nonmetropolitan	65 (9.2)	1 (0.5)	64 (13.0)	

HCAHPS indicates Hospital Consumer Assessment of Healthcare Providers and Systems; HEI, Healthcare Equality Index; IPPS, Inpatient Prospective Payment System.

satisfaction. Second, although we used HEI scores as a proxy for broader inclusion efforts at the hospital level due to limited metrics for inclusivity, they may not fully capture all dimensions of hospital inclusivity. In addition, we excluded hospitals without reported HCAHPS scores. We chose not to impute these data because nonreporting is systematically linked to data reliability, particularly for hospitals with fewer than 100 completed surveys. Imput-

TABLE 2. Direct and Indirect Associations Between Magnet Status, HEI Performance, and 2 Global Patient Satisfaction Measures

	Overall hospital rating			Likelihood to recommend the hospital		
	Estimate	95% CI	*P*	Estimate	95% CI	*P*
Average mediation (indirect) association	0.07	0.01–0.17	0.022*	0.10	0.02–0.24	0.026*
Average direct association	1.75	1.32–2.19	< 0.001**	2.37	1.77–2.98	< 0.001**
Total association	1.82	1.40–2.27	< 0.001**	2.47	1.86–3.08	< 0.001**
Proportion of mediation	0.04	0.01–0.10	0.022*	0.04	0.01–0.10	0.026*

HEI indicates Healthcare Equality Index.
*$P < 0.05$.
**$P < 0.001$.

Copyright © 2025 Wolters Kluwer Health, Inc. All rights reserved.

Copyright © 2025 Wolters Kluwer Health, Inc. Unauthorized reproduction of this article is prohibited.

TABLE 3. Direct and Indirect Associations Between Magnet Status, HEI Performance, and 6 Specific Patient Satisfaction Measures

	Nurse communication			Doctor communication			Staff responsiveness		
	Estimate	95% CI	P	Estimate	95% CI	P	Estimate	95% CI	P
Average mediation (indirect) association	0.02	−0.01–0.06	0.18	0.04	0.01–0.11	0.036*	0.06	0.01–0.14	0.046*
Average direct association	1.31	0.99–1.66	< 0.001**	0.91	0.60–1.17	< 0.001**	1.65	1.04–2.24	< 0.001**
Total association	1.33	1.00–1.68	< 0.001**	0.95	0.66–1.23	< 0.001**	1.71	1.71–1.11	< 0.001**
Proportion of mediation	0.02	−0.01–0.05	0.18	0.04	0.01–0.13	0.036*	0.03	0.01–0.09	0.046*
	Medication communication			Discharge information			Care transition		
	Estimate	95% CI	P	Estimate	95% CI	P	Estimate	95% CI	P
Average mediation (indirect) association	0.07	0.01–0.18	0.024*	0.03	−0.08–0.10	0.24	0.05	0.01–0.12	0.03*
Average direct association	1.66	1.06–2.20	< 0.001**	1.28	0.70–1.76	< 0.001**	1.48	1.16–1.91	< 0.001**
Total association	1.73	1.14–2.29	< 0.001**	1.31	0.73–1.79	< 0.001**	1.53	1.21–1.96	< 0.001**
Proportion of mediation	0.04	0.01–0.11	0.024*	0.02	−0.01–0.08	0.24	0.03	0.01–0.08	0.03*

*$P < 0.05$.
**$P < 0.001$.
HEI indicates Healthcare Equality Index.

ing such scores could introduce bias given the connection between missing data and hospital characteristics such as small patient volumes. Therefore, we addressed missing values by excluding hospitals that met CMS exclusion criteria. Another limitation is the variation in the number of patient respondents contributing to HCAHPS scores between Magnet and non-Magnet hospitals. Magnet hospitals had larger average survey samples, potentially providing more stable estimates of patient satisfaction, while smaller sample sizes in non-Magnet hospitals may increase score variability and affect comparison precision. This difference should be considered when interpreting the findings. Lastly, our findings may not fully represent smaller, resource-constrained, or for-profit hospitals that may face barriers to engaging in Magnet or HEI programs due to limited overhead and budget constraints. Future research should explore the impact of organizational inclusion strategies on patient satisfaction in a broader range of hospital settings to enhance the generalizability of the findings.

Despite these limitations, this study advances our understanding of the factors that are associated with patient satisfaction in US hospitals, highlighting the critical role of Magnet designation and hospitals' inclusion efforts in meeting the needs of diverse patient populations. Our findings suggest that fostering an inclusive hospital culture is essential for optimizing patient satisfaction.

ACKNOWLEDGMENTS

We express our gratitude to the Human Rights Campaign Foundation for generously agreeing to share their data with us.

REFERENCES

1. Manary MP, Boulding W, Staelin R, et al. The patient experience and health outcomes. *N Engl J Med*. 2013;368:201–203.
2. Lyu H, Wick EC, Housman M, et al. Patient satisfaction as a possible indicator of quality surgical care. *JAMA Surg*. 2013;148: 362–367.
3. Tsai TC, Orav EJ, Jha AK. Patient satisfaction and quality of surgical care in US hospitals. *Ann Surg*. 2015;261:2–8.
4. Centers for Medicare & Medicaid Services. The hospital Value-Based Purchasing (VBP) program. 2024. Accessed May 13, 2025. https://www.cms.gov/medicare/quality/value-based-programs/hospital-purchasing
5. Shore LM, Cleveland JN, Sanchez D. Inclusive workplaces: a review and model. *Hum Resour Manag Rev*. 2018;28:176–189.
6. Nguyen LA, Evan R, Chaudhuri S, et al. Inclusion in the workplace: an integrative literature review. *Euro J Train Dev*. 2024;48(3/4): 334–356.
7. Mor Barak ME, Lizano EL, Kim A, et al. The promise of diversity management for climate of inclusion: a state-of-the-art review and meta-analysis. *Hum Serv Organ Manag Leadersh Gov*. 2016;40:305–333.
8. Yu H, McHugh MD, Bauermeister JA, et al. LGBTQ+ inclusive policies, nurse job outcomes, and quality of care in hospitals. *JAMA Netw Open*. 2025;8:e251765.
9. Yu H, McHugh MD, Bauermeister JA, et al. The impact of hospitals' LGBTQ+ inclusion efforts on patient satisfaction from 2016 to 2023. *BMJ Qual Saf*. 2025 doi: 10.1136/bmjqs-2024-018235. Epub ahead of print.
10. Smith SA. Magnet hospitals: higher rates of patient satisfaction. *Policy Polit Nurs Pract*. 2014;15:30–41.
11. McCaughey D, McGhan GE, Rathert C, et al. Magnetic work environments: patient experience outcomes in Magnet versus non-Magnet hospitals. *Health Care Manage Rev*. 2020;45:21–31.

Copyright © 2025 Wolters Kluwer Health, Inc. All rights reserved.

Copyright © 2025 Wolters Kluwer Health, Inc. Unauthorized reproduction of this article is prohibited.

12. Stimpfel AW, Sloane DM, McHugh MD, et al. Hospitals known for nursing excellence associated with better hospital experience for patients. *Health Serv Res*. 2016;51:1120–1134.
13. Zhu J, Dy SM, Wenzel J, et al. Association of Magnet status and nurse staffing with improvements in patient experience with hospital care, 2008–2015. *Med Care*. 2018;56:111–120.
14. American Nurses Credentialing Center. Magnet model: creating a Magnet culture. 2025. Accessed May 13, 2025. https://www.nursingworld.org/organizational-programs/magnet/magnet-model/
15. Bryant T, Pruski BB, Yarbrough CB. Diversity, equity, and inclusion in the 2023 Magnet® application manual. *J Nurs Adm*. 2022;52:322–323.
16. Morgan SH. The Magnet™ model as a framework for excellence. *J Nurs Care Qual*. 2009;24:105–108.
17. Aiken LH, Sloane DM, Lake ET, et al. Organization and outcomes of inpatient AIDS care. *Med Care*. 1999;37:760–772.
18. Human Rights Campaign Foundation. Healthcare Equality Index 2024. 2024. Accessed May 13, 2025. https://www.hrc.org/resources/healthcare-equality-index
19. Centers for Medicare & Medicaid Services. Survey of patients' experiences (HCAHPS). 2024. Accessed May 13, 2025. https://data.cms.gov/provider-data/topics/hospitals/hcahps
20. Centers for Medicare & Medicaid Services. Patient survey (HCAHPS) – Hospital. 2025. Accessed May 13, 2025. https://data.cms.gov/provider-data/dataset/dgck-syfz
21. Human Rights Campaign. How do healthcare facilities near you measure up on LGBTQ+ inclusion? 2024. Accessed May 13, 2025. https://www.hrc.org/resources/healthcare-facilities
22. American Hospital Association. 2023 AHA Annual Survey. 2024. Accessed May 13, 2025. https://www.ahadata.com/system/files/media/file/2024/09/AHA-Annual-Survey-2023.pdf
23. American Nurses Credentialing Center. Find a Magnet organization. 2025. Accessed May 13, 2025. https://www.nursingworld.org/organizational-programs/magnet/find-a-magnet-organization/
24. Rodríguez-Vilá O, Nuti SV, Krumholz HM. Healthcare disparities affecting Americans in the US territories: a century-old dilemma. *Am J Med*. 2017;130:e39–e42.
25. Centers for Medicare & Medicaid Services. Technical notes for HCAHPS star ratings. 2024. Accessed May 13, 2025. https://hcahpsonline.org/globalassets/hcahps/star-ratings/tech-notes/january-2024_star-ratings_tech-notes.pdf
26. Human Rights Campaign Foundation. HEI 2024 scoring criteria. 2024. Accessed May 13, 2025. https://hrc-prod-requests.s3-us-west-2.amazonaws.com/HEI-2024-Scoring-Criteria-Tiers.pdf
27. Ayanian JZ, Weissman JS, Chasan-Taber S, et al. Quality of care for two common illnesses in teaching and nonteaching hospitals. *Health Aff*. 1998;17:194–205.
28. Hartz AJ, Krakauer H, Kuhn EM, et al. Hospital characteristics and mortality rates. *N Engl J Med*. 1989;321:1720–1725.
29. Aiken LH, Clarke SP, Sloane DM, et al. Hospital nurse staffing and patient mortality, nurse burnout, and job dissatisfaction. *JAMA*. 2002;288:1987–1993.
30. Congressional Research Service. Core Based Statistical Areas. 2024. Accessed May 13, 2025. https://sgp.fas.org/crs/misc/IF12704.pdf
31. Tingley D, Yamamoto T, Hirose Ket al. Mediation: R package for causal mediation analysis. 2014. Accessed May 13, 2025. https://cran.r-project.org/web/packages/mediation/vignettes/mediation.pdf
32. Imai K, Keele L, Tingley D, et al. Causal mediation analysis using R, . In: Vinod H, ed. *Advances in Social Science Research Using R*. Springer; 2010.
33. Robins JM, Greenland S. Identifiability and exchangeability for direct and indirect effects. *Epidemiol*. 1992;3:143–155.
34. Pearl J. Interpretation and identification of causal mediation. *Psychol Methods*. 2014;19:459–481.
35. Preacher KJ, Hayes AF. Asymptotic and resampling strategies for assessing and comparing indirect effects in multiple mediator models. *Behav Res Methods*. 2008;40:879–891.
36. Kelly LA, McHugh MD, Aiken LH. Nurse outcomes in Magnet® and non-Magnet hospitals. *J Nurs Adm*. 2011;41:428–433.
37. Rodriguez-Garcia MC, Márquez-Hernández VV, Belmonte-García T, et al. How Magnet hospital status affects nurses, patients, and organizations: a systematic review. *Am J Nurs*. 2020;120:28–38.
38. Dit Dariel OP, Regnaux J-P. Do Magnet®-accredited hospitals show improvements in nurse and patient outcomes compared to non-Magnet hospitals: a systematic review. *JBI Evid Synth*. 2015;13:168–219.

Copyright © 2025 Wolters Kluwer Health, Inc. All rights reserved.

Copyright © 2025 Wolters Kluwer Health, Inc. Unauthorized reproduction of this article is prohibited.